

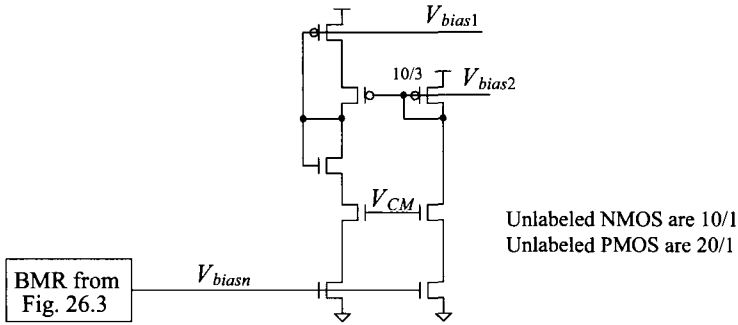


Figure 26.52 Biasing circuit for the op-amp developed in this section.

The Op-Amp's First Stage

As just mentioned, the output of the SC CMFB circuit, V_{CMFB} , moves in the same direction as movement in the average of the amplifier's outputs, v_{op} and v_{om} (the output common-mode level). Keeping this in mind, consider the bias circuit and diff-amp schematics seen in Figs. 26.52 and 26.53. We've separated the bias circuit out from the diff-amp, see Fig. 26.43, for a couple of reasons. To begin, an increase in the CMFB signal in Fig. 26.43 caused the outputs of the diff-amp to increase. As just mentioned, we want the diff-amp's outputs to move in the opposite direction of V_{CMFB} . Moving the CMFB-controlled MOSFET into the tail current of the diff-amp provides this control. Next, our bias circuit may now be shared with several op-amps. The benefit of sharing the bias circuit is a reduction in power dissipation.

Before adding the CMFB circuit, let's look at a SC circuit using an op-amp. Reviewing the sample and hold in Fig. 26.25, we see that when the ϕ switches are closed, the op-amp's inputs and outputs are at (ideally) V_{CM} . The op-amp, during this time, is in

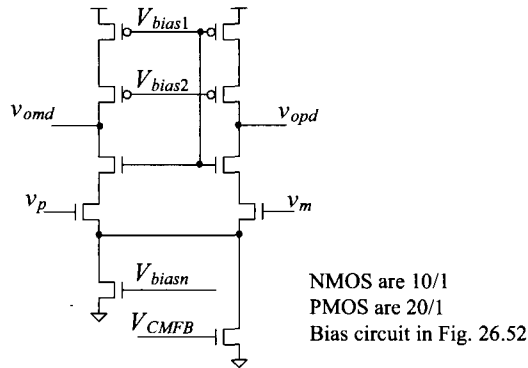


Figure 26.53 Diff-amp used with the bias circuit of Fig. 26.52.

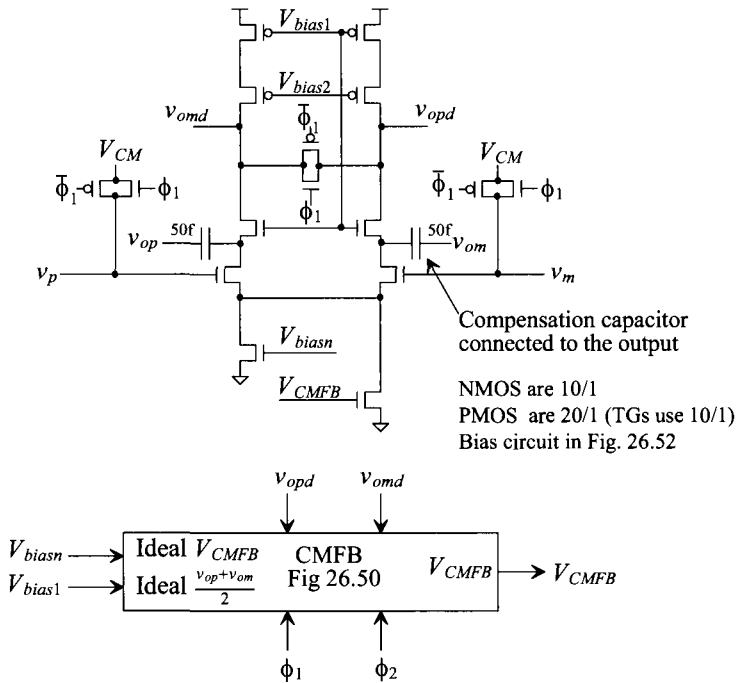


Figure 26.54 First-stage diff-amp with SC CMFB.

the unity-follower configuration. Instead of placing the op-amp in the follower configuration during this time, let's: 1) short the inputs to op-amp (Fig. 26.54) to V_{CM} , 2) short the outputs of the diff-amp (the inputs to the output buffer) together, 3) use a SC CMFB circuit around the diff-amp to ensure a balance condition (and to set the diff-amp's output voltage to V_{bias1}), 4) short the outputs of the op-amp together (so now both the output buffers inputs are shorted and outputs are shorted), and 5) use a SC CMFB around the output buffer to ensure balanced outputs. In other words, during ϕ_1 , the differential inputs are shorted together. At the same time, the outputs of the op-amp are shorted together.

Figure 26.55 shows the simulated output of the diff-amp in Fig. 26.54. The SC CMFB circuit sets the outputs of the diff-amp at V_{bias1} or roughly 600 mV. By shorting the two diff-amp outputs together and connecting the inputs to V_{CM} , we are ensuring that even with horrible mismatch, the outputs of the diff-amp will be equal and can be used to bias the output buffer. If, for example, the diff-pair shows a 50 mV mismatch (V_{OS}) then, when ϕ_1 is high, the drain currents in each side of diff-amp are set differently (making this scheme very tolerant to offsets). When ϕ_1 goes low and the op-amp is placed in a feedback configuration, each of the diff-amp's inputs will move (in opposite directions) by $V_{OS}/2$ (and so this topology doesn't store the op-amp's offset voltage).

Note that an important concern, as discussed in the last section, is the input common-mode voltage of the diff-amp. If, for example, the input common-mode voltage

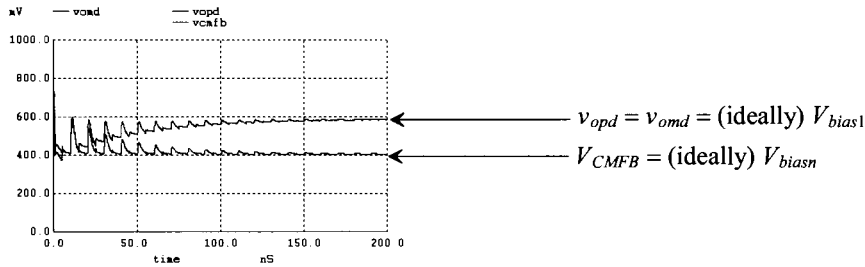


Figure 26.55 Simulating the operation of the circuit in Fig. 26.54.

drops from 500 mV, in Fig. 26.54, to 400 mV the NMOS devices will start shutting off. The op-amp will not settle or behave properly.

The Output Buffer

Figure 26.56 shows a schematic of the output buffer and SC CMFB stage. The nodes v_{omd} and v_{opd} are connected to the diff-amp outputs in Fig. 26.54. When ϕ_1 is high, the outputs of the buffer are shorted together. The SC CMFB is used to set the outputs to the common-mode voltage during this time. Figure 26.57 shows the simulation results for both stages of the op-amp (the op-amp is made with Figs. 26.54 and 26.56) with the inputs and outputs floating. When ϕ_1 is high, the inputs to the op-amp are connected to V_{CM} .

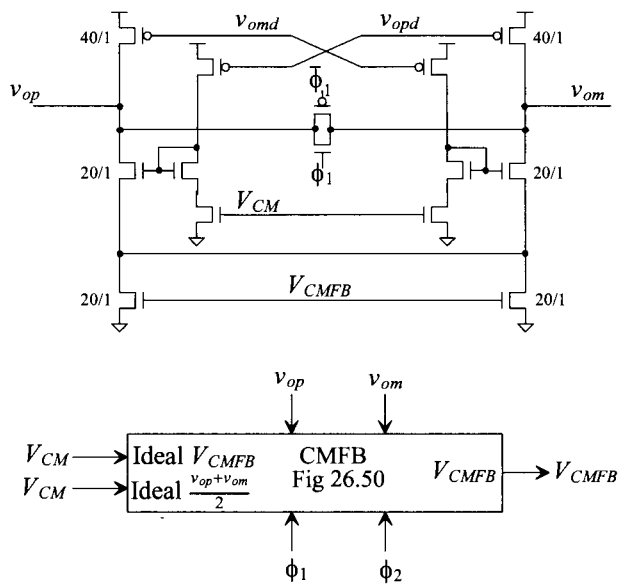


Figure 26.56 Output buffer and CMFB.

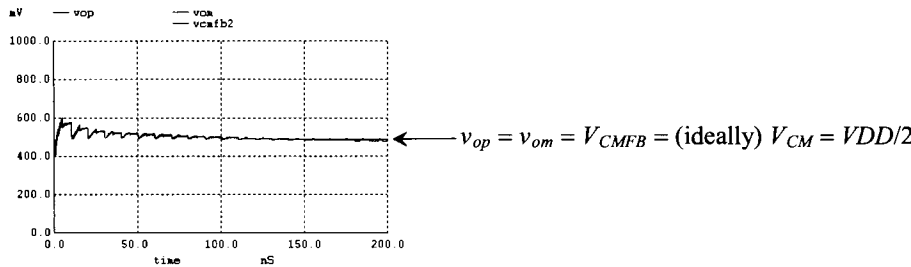


Figure 26.57 Simulating the operation of the op-amp (Figs. 26.54 and 26.56).

Is shorting the outputs of the buffer (or diff-amp) necessary? If the op-amp is operating open-loop (when ϕ_1 goes high), we don't want the outputs to float. By shorting the diff-amp and output buffer output terminals together, we ensure that they are set to a known value. If we were to use this op-amp in a SC integrator, we wouldn't short the inputs of the diff-amp to V_{cm} , the outputs of the diff-amp together, or the output buffer outputs together since there is always feedback around the op-amp. In other words, the same op-amp topology can be used in a SC integrator but without the three TGs seen in Figs. 26.54 and 26.56.

An Application of the Op-Amp

An application of the op-amp we've just developed is seen in Fig. 26.58. This is the sample and hold developed earlier (Figs. 25.15 and 26.25) except that now, during ϕ_1 (high) the inputs to the op-amp are shorted to V_{cm} (Fig. 26.54) and the outputs are shorted together (Fig. 26.56). When ϕ_2 goes high (ϕ_1 is low since the two clocks are nonoverlapping), the op-amp moves into the follower configuration and holds its outputs at the value of the inputs when the ϕ_1 switches shut off.

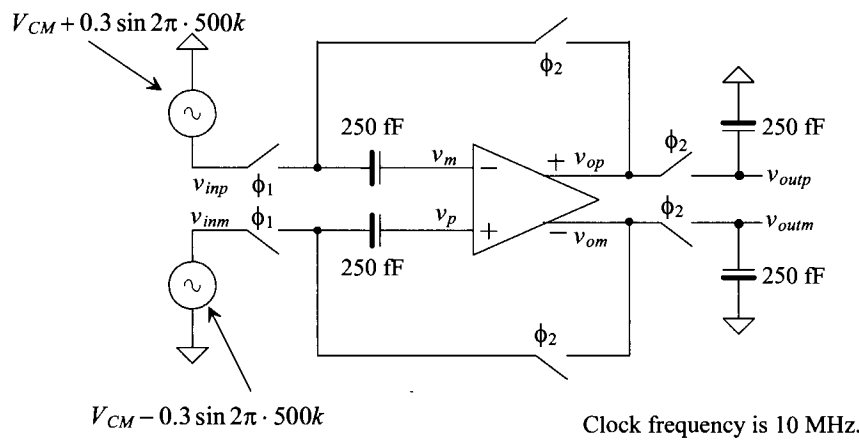


Figure 26.58 Simulating the operation of the op-amp formed with the diff-amp in Fig. 26.54 and buffer in Fig. 26.56.

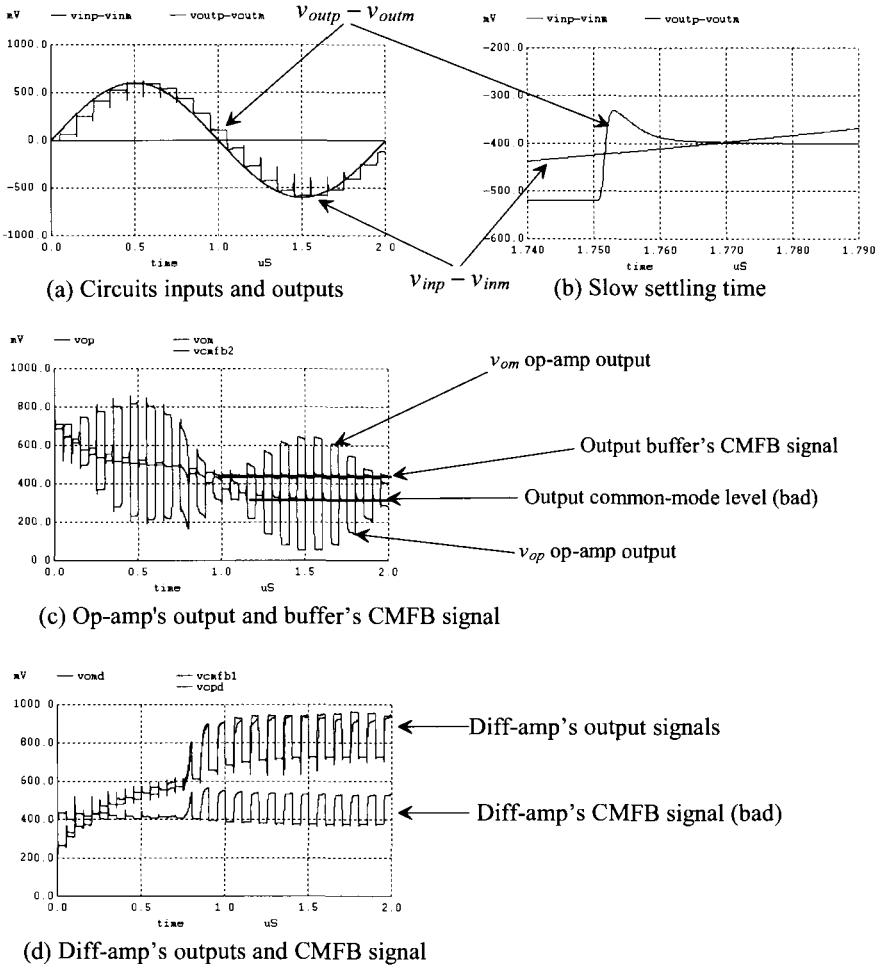


Figure 26.59 Simulating the operation of the circuit in Fig. 26.58 with the op-amp made with the diff-amp in Fig. 26.54 and the output buffer in Fig. 26.56.

Simulation Results

The simulation results are seen in Fig. 26.59. To ensure that we see the full settling behavior of the op-amp and any other issues or concerns, we use a slow clock (10 MHz) for the initial simulations. As we gain confidence that our op-amp is settling correctly and stable, we can increase the clock frequency and the input signal frequency to see how the sample and hold performs. The main thing that's different with this design compared to previous designs is that the op-amp isn't operating with feedback all of the time. When the op-amp does get put into a feedback configuration (ϕ_2 goes high), there will be a start-up time (e.g., the inputs of the op-amp will move away from V_{CM} because of the op-amp's offset).

Returning to the simulation results in Fig. 26.59, we see, in (b), that the settling time is quite long (approximately 10 ns). Letting the simulation run for a longer period of time, we see that the settling time increases and, ultimately, that the op-amp stops functioning correctly. In (c) we see the problem: the outputs of the op-amp aren't balanced around V_{CM} . Further, movement downwards causes the common-mode voltage on the input of the diff-amp to drop. The result is that the input diff-amp starts to shut off. The voltage across the diff-amp's tail current source drops and makes it difficult to control the diff-amp's output common-mode level. As seen in (d), the diff-amp's CMFB signal is increasing in an attempt to increase the current flowing in the diff-amp and pull the diff-amp's output voltages downwards. However, because the common-mode voltage on the input of the diff-amp has decreased, the voltage across the tail currents is small (tens of mV). For example, if the input common-mode voltage drops from 500 mV (V_{CM}) to 450 mV and the V_{GS} of the input diff-amp is 400 mV, then only 50 mV is left to drop across the tail current sources (one biased with V_{biasn} and the other with V_{CMFB}). We've discussed this problem earlier.

We can do one of two things to make this decrease in input common-mode voltage less of a problem: 1) reduce the diff-amp's bias current or 2) increase the widths of the input diff-pair. In both cases we need to increase the voltage dropped across the tail current sources by decreasing the V_{GS} of the NMOS diff-pair (the inputs to the diff-amp are, ideally, at V_{CM}). If we decrease the bias current, the g_m of the diff-pair decreases and so does the speed of the op-amp (remembering $f_{un} = g_m/2\pi C_c$). By increasing the width of the diff-pair, we increase g_m and thus the unity-gain frequency of the op-amp. The issue with increasing the width of the diff-pair is that the overdrive voltage for these MOSFETs decreases and thus so does their f_T . If we keep the increase in the width to a modest level, the parasitic pole associated with these diff-pair MOSFETs shouldn't, in any significant way, affect the stability of the op-amp.

Figure 26.60 shows the operation of the op-amp made with the diff-amp in Fig. 26.54 and the output buffer in Fig. 26.56 if we increase the widths of the diff-pair from 10 to 40. We also changed the MOSFETs in the biasing circuit Fig. 26.52 with gates connected to V_{CM} to have widths of 40 to maintain symmetry (important for reducing the offset voltage). To further increase the speed we also reduced C_c by 4 (from 50 to 12.5 fF) and increased the widths of the devices in the output buffer by 4 (to move f_2 up and away from f_{un}). The settling time, as seen in (b), is < 3 ns. Note that increasing the g_m of the output buffer by using wider devices (more current) increases f_2 allowing a corresponding increase in f_{un} . This is an **important** technique for increasing the speed of an op-amp.

Reviewing the op-amp designs in the last section, we see that their performance may also be improved by a modest increase in the widths of the diff-pair and output buffer. However, as just mentioned, we have to be careful to maintain symmetry. For example, if we take the op-amp in Fig. 26.43, increase the widths of the first stage diff-pair from 10 to 40, and simulate, we see that the outputs don't swing all the way up (or down) to the correct values. Further investigation reveals that the diff-pair moved into the triode region. To move the MOSFETs back into the saturation region, we must also increase the widths of the MOSFETs in the bias circuit (with gates at V_{CM}) from 10 to 40. This adjusts the bias voltages for the NMOS/PMOS diff-amp loads. By providing the CMFB back through the diff-amp (instead of through the output buffer), we can adjust the diff-amp's output voltage. Thus the op-amp is more tolerant to issues in the bias circuit.

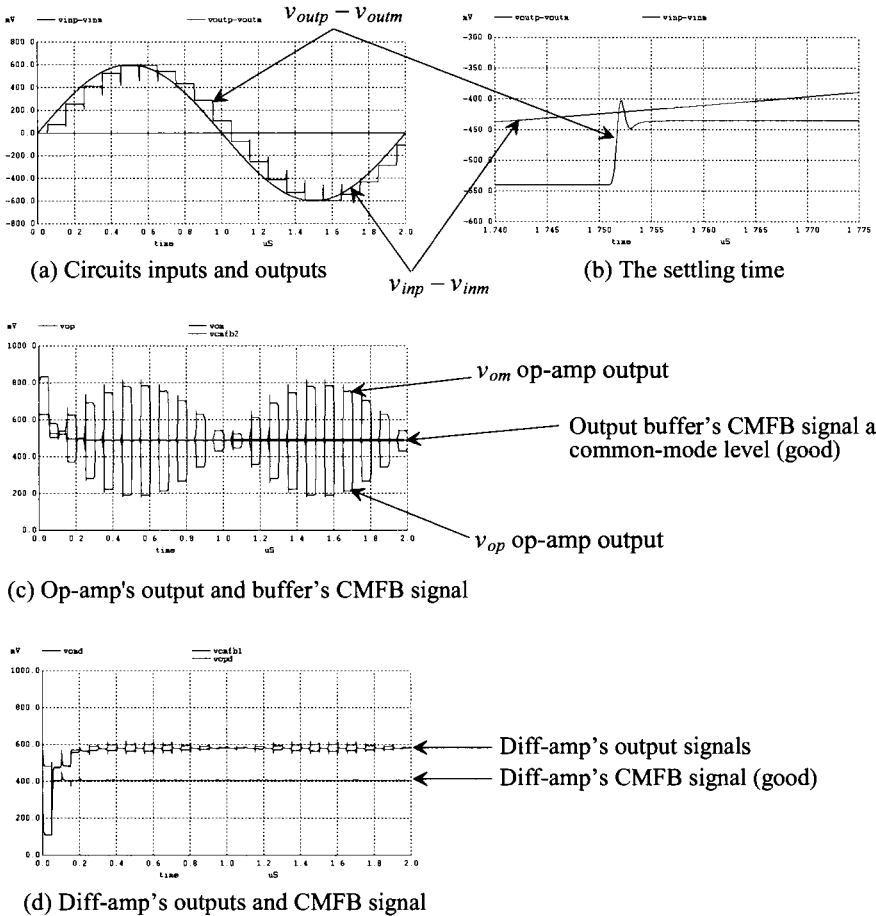


Figure 26.60 Regenerating the data in Fig. 26.59 after increasing the widths of the input diff-pair and output buffer by 4 while reducing the compensation capacitors by 4 (from 50 to 12.5 fF).

A Final Note Concerning Biasing

In (most) of the designs presented in this chapter, we biased the gates of the NMOS cascode devices in the diff-amp's load at the same (ideally) voltage as the diff-amp's output. For example, in Fig. 26.53, the gate of the NMOS device is tied to V_{bias1} , while its drain (the output of the diff-amp) is also (ideally) at V_{bias1} . This selection is fine for learning design. It minimizes power by avoiding an extra bias reference circuit. However, if VDD starts to drop the drain-source voltages across the diff-pair and tail current sources can drop to the point where the op-amp shuts off. Again, increasing the widths of the diff-pair helps with this concern. Also, VDD noise can be more of a concern since we usually want the NMOS devices' gates referenced to ground (V_{bias1} is referenced to VDD).

A more general, and better, solution is to bias the gates of the NMOS with a gate-drain-connected NMOS, as seen in Figs. 22.30, 26.62, and 26.63. This allows the gate voltage of the NMOS to move to a higher voltage than its drain (allowing the

op-amp to function with lower V_{DD}). For example, the diff-amp in Fig. 26.53 has a DC output voltage of roughly 600 mV ($= V_{bias1}$). A more appropriate voltage for the gate of the NMOS device is 800 mV. Remembering that the gate-source voltages of the NMOS diff-pair (Fig. 26.18) are roughly 400 mV puts the drains of the diff-pair at 400 mV. The result: V_{DD} is more evenly divided across the devices in the diff-amp.

ADDITIONAL READING

- [1] V. S. L. Cheung, H. C. Luong, M. Chan, and W. H. Ki, "A 1-V 3.5mW CMOS Switched-Opamp Quadrature IF Circuitry for Bluetooth Receivers," *Symp. VLSI Circuits Dig.* 16, pp. 140–143, June 2002.
- [2] M. Keskin, U. Moon, and G. C. Temes, "A 1-V 10-MHz clock-rate 13-bit CMOS DS modulator using unity-gain-reset opamps," *IEEE Journal of Solid-State Circuits*, vol. 37, pp. 817–824, July 2002.
- [3] V. S. L. Cheung, H. C. Luong, and W. H. Ki, "A 1V CMOS switched-opamp switched-capacitor pseudo-2-path filter," *IEEE International Solid-State Circuits Conference*, vol. 35, pp. 154–155, February 2000.
- [4] I. E. Opris, L. D. Lewicki, and B. C. Wong, "A single-ended 12-bit 20 Msample/s self-calibrating pipeline A/D converter," *IEEE Journal of Solid-State Circuits*, vol. 33, pp. 1898–1903, December 1998.
- [5] D. C. Thelen Jr. and D. D. Chu, "A low noise readout detector circuit for nanoampere sensor applications," *IEEE Journal of Solid-State Circuits*, vol. 32, pp. 337–348, March 1997.
- [6] P. D. Walker and M. M. Green, "A tuneable pulse-shaping filter for use in a nuclear spectrometer system," *IEEE Journal of Solid-State Circuits*, vol. 31, pp. 850–855, June 1996.
- [7] G. Caiulo, F. Maloberti, G. Palmisano, and S. Portaluri, "Video CMOS power buffer with extended linearity," *IEEE Journal of Solid-State Circuits*, vol. 28, pp. 845–848, July 1993.
- [8] K. Nakamura and L. R. Carley, "An enhanced fully differential folded-cascode op amp," *IEEE Journal of Solid-State Circuits*, vol. 27, pp. 563–568, April 1992.
- [9] R. Castello, G. Nicollini, and P. Monguzzi, "A high-linearity 50- Ω CMOS differential driver for ISDN applications," *IEEE Journal of Solid-State Circuits*, vol. 26, pp. 1809–1816, December 1991.
- [10] S. M. Mallya and J. H. Nevin, "Design procedures for a fully differential folded-cascode CMOS operational amplifier," *IEEE Journal of Solid-State Circuits*, vol. 24, pp. 1737–1740, December 1989.
- [11] M. Banu, J. M. Khoury, and Y. Tsividis, "Fully Differential Operational Amplifiers with Accurate Output Balancing," *IEEE Journal of Solid State Circuits*, vol. 23, No. 6, pp. 1410–1414, December 1988.
- [12] R. Castello and P. R. Gray, "A high-performance micropower switched-capacitor filter," *IEEE Journal of Solid-State Circuits*, vol. 20, pp. 1122–1132, Dec. 1985.

- [13] T. C. Choi, R. T. Kaneshiro, R. Broderson, and P. R. Gray, "High-Frequency CMOS Switched Capacitor Filters for Communication Applications," *IEEE Journal of Solid State Circuits*, vol. SC-18, pp. 652–664, December 1983.
- [14] D. Senderowicz, S. F. Dreyer, J. H. Huggins, C. F. Rahim, and C. A. Laber, "A family of differential NMOS analog circuits for a PCM codec filter chip," *IEEE Journal of Solid-State Circuits*, vol. 17, pp. 1014–1023, December 1982.

PROBLEMS

Unless otherwise indicated, use the 50 nm CMOS process from the examples in this chapter, the biasing circuit in Fig. 26.3, and 10/1 NMOS and 20/1 PMOS.

- 26.1** Using simulations, determine the transition frequencies, f_T , for the NMOS and PMOS devices seen in Fig. 26.1 at the nominal operating conditions indicated in the figure. Show that by increasing the MOSFET's overdrive voltage, the f_T s of the MOSFETs increases.
- 26.2** Simulate the operation of the two-stage op-amp in Fig. 26.2. Show that the quiescent current in the output buffers is considerably below the desired 20 μ A. (Note that the inputs of the op-amp should be held at V_{CM} in the simulation to keep the diff-amp conducting current.) Does this affect the speed of the output buffer? Why or why not?
- 26.3** Plot reference current against resistor value for the BMR seen in Fig. 26.3. Use simulations to determine the I_{REF} for each value of resistance.
- 26.4** Comment on the benefits and/or concerns with the CMFB input connections seen in Fig. 26.61. Use simulations to support your answers. Note the similarity to the way CMFB was implemented in Fig. 26.17.

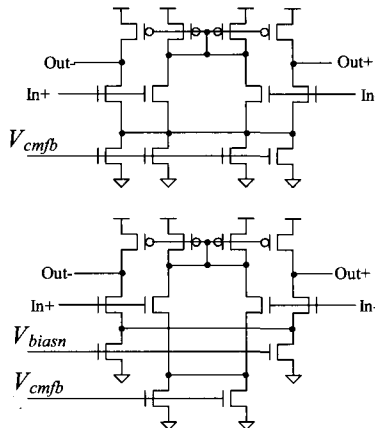


Figure 26.61 Circuits for Problem 26.4.

- 26.5** Verify, using simulations, that the circuit in Fig. 26.9 does indeed amplify the difference between V_{biasp} and the average on the + inputs of the amplifier.

Comment on the operation of the circuit, making sure it is clear that the limitations, uses, and operation of the amplifier are understood.

- 26.6** Simulate the operation of the CMFB circuit in Fig. 26.16.
- 26.7** Is V_{DD} divided evenly amongst the drain-source voltages of the MOSFETs in Fig. 26.19? Suggest a method to better divide V_{DD} amongst the MOSFETs used in the diff-amp.
- 26.8** Suggest a simple method to speed up the step response of the op-amp in Fig. 26.22 in the circuit of Fig. 26.24. Verify the validity of your suggestion using SPICE simulations.
- 26.9** Suggest an alternative method to the one seen in Fig. 26.32 for controlling the output buffer's current. Using the modification, regenerate the results seen in Fig. 26.36.
- 26.10** Is the CMFB loop stable in the op-amp of Fig. 26.40? Use the large-signal test circuit seen in Fig. 26.24 (at a slower frequency) and simulations to look at the stability of this loop. Suggest, and verify with simulations, methods to improve the stability of the CMFB loop.
- 26.11** Repeat Problem 26.10 for the op-amp in Fig. 26.43.
- 26.12** As discussed at the end of the chapter and in Fig. 26.59 and the associated discussion, the input common-mode voltage which drops below V_{CM} , can shut off the op-amp's input diff-amp and cause problems. The diff-amp's NMOS devices have a nominal V_{GS} of 400 mV, leaving only 100 mV across the diff-amp's tail current. Show, using the op-amp in Fig. 26.39 in the configuration seen in Fig. 26.24 with an input common-mode voltage less than 500 mV (the input pulse waveforms average to a voltage less than V_{CM}), the resulting problems. Show that increasing the widths of the NMOS diff-pair (and the mirrored NMOS in the bias circuit with gates tied to V_{CM}) from 10 to 30 helps to increase the operating range (four MOSFET widths are increased from 10 to 30). What happens if only the diff-pair widths (two MOSFETs) are increased?
- 26.13** Repeat Problem 26.12 for the op-amp in Fig. 26.40. What happens if the input common-mode voltage becomes greater than 500 mV?
- 26.14** Repeat Problem 26.12 for the op-amp in Fig. 26.43. What happens if the input common-mode voltage becomes greater than 500 mV? Why does the op-amp in Fig. 26.40 perform better with variations in the input common-mode voltage than the op-amp in Fig. 26.43?
- 26.15** Design and simulate the operation of an op-amp using gain-enhancement (Fig. 26.42) and with an open-loop DC gain greater than 2,000, based on the topologies seen in Figs. 26.40 or 26.43. Simulate the operation of your design and generate outputs like those seen in Fig. 26.44.
- 26.16** Figure 26.62 shows an op-amp based on the topology seen in Fig. 26.40 but biased for lower V_{DD} operation. Select the size of the added gate-drain connected MOSFET to allow for proper operation. Simulate the operation of the design showing the DC gain and large signal step responses (as in Fig. 26.34 and 26.35).

- 26.17** Repeat Problem 26.16 for the op-amp in Fig. 26.43.
- 26.18** Using the diff-amp in Fig. 26.63 in the configuration seen in Fig. 26.54 (with SC CMFB) and the output buffer in Fig. 26.56 (again with SC CMFB), regenerate the waveforms seen in Fig. 26.60.
- 26.19** A switched-capacitor integrator is an example of a circuit that uses an op-amp that can't have the outputs of its diff-amp or output buffer shorted when ϕ_1 goes high. Using the diff-amp in Fig. 26.63 (with SC CMFB) and the output buffer in Fig. 26.56 without the switches (again with SC CMFB), demonstrate the operation of a SC integrator.

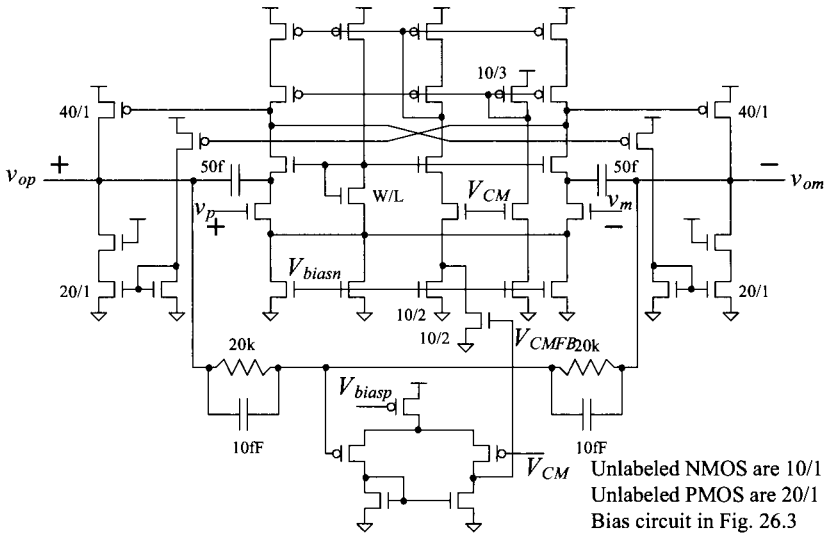


Figure 26.62 An op-amp with better biasing (but more power) for lower VDD operation.

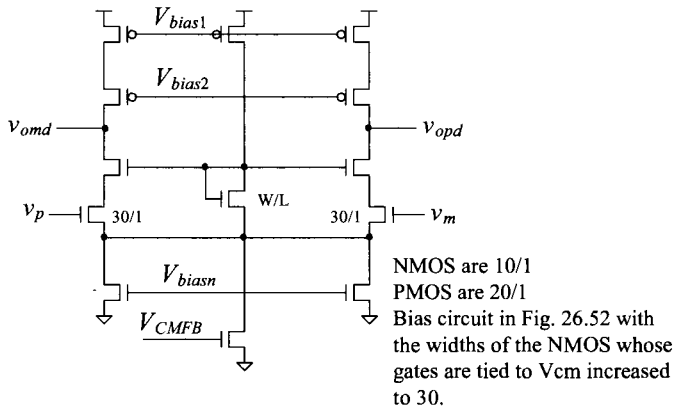


Figure 26.63 Modifying the diff-amp seen in Fig. 26.53 for wider swing operation.